mfrmr quick reference

mfrmr package

Data to fitted results

A compact reference for the data review / fit / summary / plot / diagnose / report loop in mfrmr. Every snippet assumes `library(mfrmr)` and a long-format `data.frame` with `Person`, one or more facet columns, and `Score`.

1. Check data roles, score support, and missing codes

```
toy <- load_mfrmr_data("example_operational")

data_review <- describe_mfrm_data(
  toy, person = "Person", facets = c("Rater", "Criterion"),
  score = "Score", rating_min = 1, rating_max = 4
)
summary(data_review)

# Declare only codes that mean missing in the study.
dirty_review <- describe_mfrm_data(
  dirty_data, "Person", c("Rater", "Criterion"), "Score",
  missing_codes = c(99, 999, -1, "N/A", "")
)
```

Confirm retained and zero-count categories and inspect the score map. Use the same `rating_min`, `rating_max`, `keep_original`, and `missing_codes` decisions in `fit_mfrm()`.

2. Fit

```r
fit <- fit_mfrm(
  toy,
  person = "Person",
  facets = c("Rater", "Criterion"),
  score  = "Score",
  method = "MML",           # recommended starting method
  model  = "RSM",           # use PCM/GPCM only with a model rationale
  rating_min = 1,
  rating_max = 4
)
```

For PCM, set the step facet explicitly when its role is not unambiguous. Treat bounded GPCM as a slope-aware sensitivity extension and read `gpcm_capability_matrix()` before using downstream helpers.

3. Read summaries, then draw the required Wright map

```r
fit_summary <- summary(fit, profile = "fit", detail = "brief")

# Comprehensive FACETS-organized review; mfrmr remains the estimator.
review <- summary(fit, profile = "facets", detail = "brief")
res <- review$results
diag <- res$diagnostics

# Primary final-scale display: all locations plus available facet uncertainty.
plot(res, type = "wright", renderer = "native",
     show_ci = TRUE, top_n = Inf, preset = "publication")
```

Read optimizer status and terminal-gradient guidance before interpreting estimates. The native map is the primary result because it keeps facet uncertainty and fitted step locations visible.

4. Add only the diagnostics needed for the question

```r
# Reuse review diagnostics; rerun only for explicit custom settings.
diag_pca <- diagnose_mfrm(fit, residual_pca = "both")

# Optional follow-up: x = Infit, y = measure. Persons are opt-in.
plot(fit, type = "fit_pathway", diagnostics = diag,
     fit_stat = "Infit", include_person = FALSE)
```

5. Report / APA

```r
chk  <- reporting_checklist(fit, diagnostics = diag)
apa  <- build_apa_outputs(fit, diag)
cat(apa$report_text)

tbl  <- apa_table(fit, which = "facets", diagnostics = diag)
kab  <- as_kable(tbl)       # Markdown / HTML via kableExtra
ft   <- as_flextable(tbl)   # Word / PowerPoint via flextable
```

6. Bias / DFF / Equivalence

```r
bias <- estimate_bias(fit, diag, facet_a = "Rater", facet_b = "Criterion")
summary(bias)
plot_bias_interaction(bias, plot = "ranked", show_ci = TRUE)

# Requires a study data frame containing the named grouping column.
dff <- analyze_dff(fit, diagnostics = diag,
                   facet = "Rater", group = "Group", data = your_data)
plot_dif_summary(dff)

# Requires inference-ready MML and unregularized joint covariance.
# Choose the practical bound before inspecting equivalence results.
eq <- analyze_facet_equivalence(fit, facet = "Rater",
                                equivalence_bound = 0.5, ci_level = 0.95)
plot_facet_equivalence(eq, type = "forest")
```

7. Hierarchical structure and small-N review

```r
review <- facet_small_sample_review(fit)
summary(review)

icc <- compute_facet_icc(
  toy, facets = c("Rater", "Criterion"),
  score = "Score", person = "Person",
  ci_method = "profile"      # "boot" for parametric bootstrap CI
)

h <- analyze_hierarchical_structure(
  toy, facets = c("Rater", "Criterion"),
  ci_method = "profile"
)
```

8. Empirical-Bayes shrinkage for small-N facets

```r
# Integrated:
fit_eb <- fit_mfrm(toy, "Person", c("Rater", "Criterion"), "Score",
                   method = "MML", facet_shrinkage = "empirical_bayes")
shrinkage_report(fit_eb)

# Post-hoc:
fit_eb2 <- apply_empirical_bayes_shrinkage(fit)
```

9. Plot surface (selected)

Function	Purpose
`plot(fit)`	Wright map (default)
`plot_compare_mfrm(fit_rsm, fit_pcm, type = "ccc", view = "difference")`	Conditional probability differences; paired Wright views also available
`plot_wright_unified(fit, renderer="facets")`	Optional FACETS-style `*` ruler; visual convention only
`plot(fit, type="fit_pathway", diagnostics=diag, include_person=TRUE)`	Infit-vs-measure pathway; person rows opt-in
`plot_threshold_ladder(fit)`	Category threshold ordering
`plot_person_fit(fit, diag)`	Per-person Infit/Outfit bubble
`plot_rater_severity_profile(fit, diag)`	Rater ranking + CI bands
`plot_bias_interaction(bias, plot="heatmap")`	Bias cell heatmap
`plot_displacement(fit, show_ci=TRUE)`	Anchor displacement lollipop
`plot_fair_average(fit, show_ci=TRUE)`	Fair-score diagnostic intervals; full-refit coverage unverified
`plot_anchor_drift(drift, type="forest")`	Wave-level anchor CI forest
`plot(chain, type="graph")`	Wave/common-element graph with per-link screening
`plot(chain, type="links")`	Recorded wave links; counts of retained elements
`plot(chain, type="anchor_removal")`	Newly disconnected wave pairs after each common-element deletion; no refitting
`plot(chain, type="offset_sensitivity")`	Rescreened offset changes after deletion; source estimates/SEs fixed

10. Reproducibility and export

```r
# The default warning reminds you that this does not deidentify the files.
export_mfrm_bundle(fit, diagnostics = diag,
                   output_dir = "mfrmr_bundle/")
export_mfrm_results(res, output_dir = "mfrmr-results",
                    preset = "starter", overwrite = TRUE)
# Add acknowledge_sensitive = TRUE only after reviewing every intended file
# and confirming that the destination follows the study's handling policy.
manifest <- build_mfrm_manifest(fit, diagnostics = diag)
# manifest$environment, $dependencies, $input_summary, $session_info,
# $hierarchical_review, $missing_recoding, $shrinkage_review
replay_script <- build_mfrm_replay_script(fit, path = "replay_mfrmr.R")
```

11. FACETS and ConQuest users

- Keep the native Wright map with available facet uncertainty as the primary mfrmr display. Use `renderer = "facets"` when readers benefit from FACETS-style asterisks and ruler layout; this does not reproduce FACETS estimation.

```r
rubric_labels <- setNames(
  your_rubric_labels,
  fit$prep$score_map$OriginalScore
)
plot_wright_unified(
  fit, renderer = "facets", show_ci = FALSE,
  category_labels = rubric_labels
)
```

- Use `facets_feature_coverage()` for feature-by-feature alternatives.
- Use `build_conquest_overlap_bundle(fit_lr)` only after fitting the documented unidimensional MML comparison. Run the generated `.cqc` file in ConQuest, then pass its four comparison CSV exports to `normalize_conquest_overlap_exports()` and `review_conquest_overlap()`; retain the additional history CSV for objective/free-dimension verification.
- mfrmr does not execute FACETS or ConQuest and does not claim general numerical equivalence with either program.

Package links

- Source and documentation: https://github.com/Ryuya-dot-com/mfrmr
- Citation: `citation("mfrmr")`